

CUSTOMER PURCHASE PATTERN ANALYZER

Business Analysis Document

Project Objective

The objective of this project is to analyze customer purchasing behavior and identify patterns that can help businesses improve sales performance, customer retention, product recommendations, and marketing effectiveness.

Dataset Overview

Dataset Name

Customer Purchase Pattern Analyzer Dataset

Total Records

120 Customer Transactions

Total Attributes

20 Columns

Key Fields

- Customer ID
- Customer Name
- Age
- Gender
- City
- Region
- Occupation
- Product Category
- Product Name
- Purchase Date
- Quantity Purchased
- Unit Price
- Total Purchase Value
- Payment Method
- Purchase Channel
- Customer Segment
- Loyalty Status

- Discount Used
 - Purchase Frequency
 - Last Purchase Date
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Data Cleaning Activities

Missing Value Handling

- Checked all columns for null values
- Replaced missing categorical values using mode
- Replaced missing numerical values using median

Duplicate Removal

- Removed duplicate transaction records
- Ensured Customer IDs remained unique

Date Formatting

- Converted Purchase Date into Date format
- Converted Last Purchase Date into Date format

Data Validation

- Checked negative sales values
- Verified quantity values
- Ensured revenue calculations were correct

Category Standardization

- Standardized customer segments
 - Standardized loyalty categories
 - Standardized payment methods
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Feature Engineering

Created additional business metrics:

Revenue

Revenue = Quantity Purchased × Unit Price

Customer Age Groups

- 18–25
- 26–35
- 36–45
- 46–55
- 56+

Purchase Recency

Difference between current date and last purchase date.

Customer Value Classification

- High Value
 - Medium Value
 - Low Value
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Key Performance Indicators

Total Revenue

Total income generated from all purchases.

Total Customers

Unique customers in the dataset.

Total Orders

Total purchase transactions.

Average Order Value (AOV)

$AOV = \text{Total Revenue} / \text{Total Orders}$

Repeat Purchase Rate

$\text{Repeat Purchase Rate} = \text{Repeat Customers} / \text{Total Customers} \times 100$

Customer Lifetime Value (CLV)

$CLV = \text{Average Order Value} \times \text{Purchase Frequency}$

Customer Segmentation Analysis

Premium Customers

Highest revenue contributors.

Regular Customers

Moderate spending behavior.

Budget Customers

Price-sensitive customers.

Loyalty Analysis

Gold Customers: Most profitable segment.

Silver Customers: Moderate engagement.

Bronze Customers: Potential growth opportunities.

Non-Members: Target group for loyalty campaigns.

Product Performance Analysis

Best-Selling Categories

- Electronics
- Fashion
- Grocery

Revenue Contribution

Identified categories generating maximum sales.

Product Popularity

Measured using purchase frequency.

Regional Analysis

Analyzed sales performance across:

- North
- South
- East
- West

Identified high-performing regions and low-performing markets.

Payment Method Analysis

Most used methods:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Cash

Purpose: Understand customer payment preferences.

Purchase Channel Analysis

Channels analyzed:

- Online
- Mobile App
- Physical Store

Insights generated regarding preferred purchasing channels.

Customer Retention Analysis

Identified:

- Repeat Customers
- One-Time Buyers
- Loyal Customers
- At-Risk Customers

Recommended retention strategies based on purchase frequency.

Business Insights

Insight 1

Premium customers contribute the highest revenue.

Insight 2

Electronics category generates maximum sales.

Insight 3

Online purchases dominate customer transactions.

Insight 4

Gold loyalty members show the highest retention rate.

Insight 5

Customers aged 26–35 contribute the largest share of revenue.

Insight 6

UPI is the preferred payment method.

Insight 7

Repeat customers generate significantly more revenue than one-time buyers.

Recommendations

1. Strengthen loyalty programs.
 2. Offer personalized promotions.
 3. Target high-value customers with exclusive offers.
 4. Improve retention campaigns.
 5. Cross-sell frequently purchased products.
 6. Focus marketing efforts on high-performing regions.
 7. Encourage repeat purchases through reward systems.
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Conclusion

The Customer Purchase Pattern Analyzer successfully identified customer buying trends, purchasing preferences, high-value customer groups, profitable product categories, and customer retention opportunities. The insights generated can help businesses improve customer engagement, maximize revenue, and enhance overall business performance.